- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. Unless otherwise instructed, when a numerical value is needed, use $g = 9.8$.

9 A cannonball is fired from a point on horizontal ground at $100\,\mathrm{m\,s^{-1}}$ at an angle of 25° above the horizontal. Ignoring air resistance, calculate

(a) the greatest height the cannonball reaches, **[3]**

(b) the range of the cannonball. **[4]**